CEZERİFEST 2026

3D DESIGN AND PRINTING COMPETITION

SUBMISSION FILE

Participation Type

Please select one:

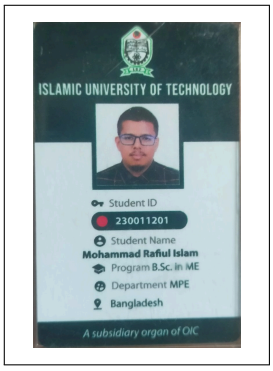
- ☐ Individual Participation
- ☑ Group Participation (Max 2 members)

For Group Participation

Group Member 1 Information

- **Full Name:** Mohammad Rafiul Islam
- **Email Address:** rafiulislam2220@gmail.com
- **Phone Number:** 01814924640
- **Address:** Perchivil Hill, Chawkbazar, Chittagong
- **Country:** Bangladesh
- **Educational Background:** Islamic University of Technology (IUT), OIC





Group Member 2 Information

- **Full Name:** Tasnia Basharat Mahi
- **Email Address:** tasniabasharat@gmail.com
- **Phone Number:** 01671012856
- **Address:** Rajpara, Tilaghar, Sylhet, Bangladesh.
- **Country:** Bangladesh
- **Educational Background:** Islamic University of Technology (IUT), OIC





1- Project Information

Project Title: ZeroWatt Wind

Project Objective:

i. Design and prototype a passive, wind-driven mechanical system requiring no electricity

ii. Capture ambient wind energy using a Fibonacci-spiral turbine, optimized for low, inconsistent urban wind speeds

iii. Transmit that rotational energy via a gear drivetrain from the point of capture to the point of need

iv. Provide a reliable, low-cost alternative to electric HVAC and aerators during power outages and in dense urban or low-income areas

Participant Organisation:

Both participants are studying Mechanical Engineering at IUT and have a special interest in fluid mechanics.

The Project's Contribution to the City:

Application 1: Passive Wind Power Harnessing Air Ventilator

The Urban Problem: To keep outdoor pollution and rising city heat out, new buildings are built airtight. Thick insulated walls, sealed windows and no natural airflow gaps. While that is good when the purpose is to keep the bad air out, it also means the building can no longer breathe on its own. Every bit of fresh air and cooling now has to come from electric HVAC systems, which during load shedding, cannot function.

Reason why Passive Ventilation can't compensate: Cities have traditionally used passive ventilation (such as windows and basic thermal buoyancy vents) but these are largely insufficient in high density urban environments, being too weak to overcome the internal heat loads generated by people and electronics.

Affects on people: Stagnant air leads to poor indoor air quality, making occupants sick (known as Sick Building Syndrome). CO_2 builds up as exhaled air isn't cleared out. Harvard's COGfx study found a 500 ppm CO_2 rise cut cognitive performance by 15%, with higher $PM_{2.5}$ also slowing response times and accuracy.

Solution: Our designed system directly targets this issue. Wind speed increases significantly with height due to reduced surface friction, which is why a Fibonacci turbine is intentionally mounted at rooftop level to maximize wind energy capture. It drives the motion through a two-stage bevel gear system, redirecting rotation 90° to walk the motion down through the wall or ceiling cavity to a zone that would otherwise see little to no natural airflow. It is in the upper cavity where the system intercepts the natural phenomenon of thermal buoyancy. In any enclosed space, heat generated by occupants and electronics naturally rises and pools at the ceiling. While passive vents are often too weak to clear this heavy heat load, our system actively drives NACA 65-810 aerodynamically efficient fan blades to aggressively force this buoyant hot air out. By actively exhausting this upper heat layer, it creates a negative pressure that draws fresh, cooler air into the lower occupant zones As a result, we get a fully passive system with no failure during load shedding and no added carbon footprint.

The moving air also continuously carries humidity out and replaces it with drier air, which prevents the damp conditions that cause mould growth and wood rot.





Application 2: Pond Aerator to increase the amount of Dissolved Oxygen

Problem: In a rapidly developing city, urban water bodies face a silently growing ecological threat, that is stagnation. Surrounded by concrete, these bodies of water are cut off from natural currents. Without continuous movement, they lose dissolved oxygen, which leads to toxic algae blooms, foul odors and ideal breeding grounds for mosquitoes. This, in turn, disrupts aquatic habitats.

Traditional Approach: Traditional answer to it is dropping electrically powered aerators into the water. But running underwater high-voltage electrical cables across public parks is dangerous, expensive to install and creates a permanent drain on the city's power grid.

Solution: The same mechanism we designed solves this very problem efficiently. Our system uses a surface paddle wheel. From an engineering standpoint, this is the most efficient way to increase dissolved oxygen (DO). The rotating paddles aggressively break the surface tension of the water, physically splashing and churning it. This maximizes the surface area of the water exposed to the air, accelerating natural gas exchange, while simultaneously destroying the still-water surface tension. All while it is powered by wind, instead of electricity.

Unlike a constantly running electric aerator, our system doesn't need to run continuously to be effective. Pond stagnation develops due to thermal stratification where warm surface water sits on cooler, oxygen-poor water below. Intermittent aeration, triggered naturally by gusts of wind, is enough to repeatedly disrupt this layering before it stabilizes. In long periods without wind, the pond behaves similarly to any water body with limited wind access but the wind conditions in most urban open spaces are rarely still for extended stretches and even brief aeration events meaningfully slow the rate of oxygen depletion.



Innovation offered by the Project:

Our turbine uses a fibonacci (logarithmic spiral) blade arrangement, where blade pitch and spacing follow the golden-ratio progression found throughout nature (nautilus shells, sunflower seed heads, pinecones). Some of its features are as follows:

i. **Progressive pitch:** each blade meets the wind at a gradually changing angle along its spiral path, which smooths airflow entry and reduces the abrupt separation that causes turbulence and drag in flat bladed rotors.

ii. **Self-starting at low wind speeds:** the asymmetric spacing means at any given moment, several blades are positioned at different effective angles of attack, improving the odds that at least one blade catches enough torque to initiate rotation even in light or gusty urban wind, a known weak point of a ventilator known as whirlybirds.

iii. **Reduced vibration and noise:** It creates quieter, more consistent rotation, which matters for a device meant to sit above occupied rooms or public parks.

For the fan's blade part, we moved away from flat blades entirely and selected the NACA 65-810 airfoil profile, a blade originally developed for axial-flow fans and compressors. It's a proven, published profile engineered for low-speed, high-efficiency air movement. We wrote airfoil shape equation using scipy in python which helped to create NACA 65-810 shape more accurately. Some of the features are:

i. **Lift-based airflow:** Cambered cross-section moves more air per rotation than a flat blade,

for the same input torque.

ii. **Compensates for gear losses:** Higher aerodynamic efficiency offsets torque lost through the two-stage bevel gear drivetrain.

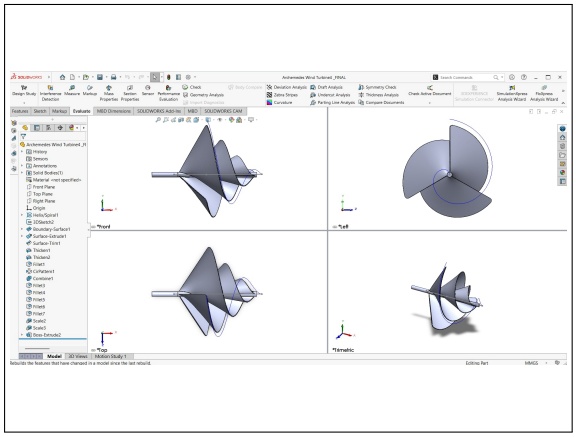
iii. **Low turbulence:** Minimizes flow separation at low fan speeds, producing a steady breeze instead of choppy gusts.

iv. **Quieter operation:** Smoother airflow transition reduces the noise typically caused by turbulent, flat-blade fans.

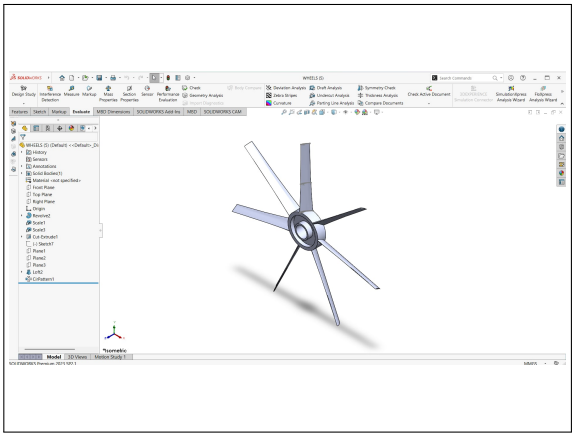
Project Specifications:

Material: PLA
Design Software: SolidWorks
3D Printer: Bambulab p1s

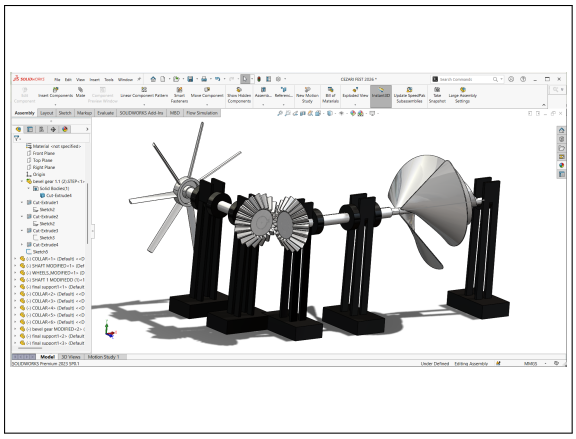
Project's 3D Design Phase:



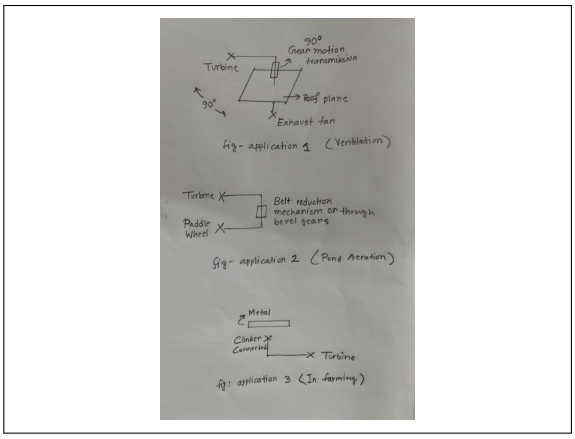
Archimedes Turbine



NACA 65-810 Rotor Blade



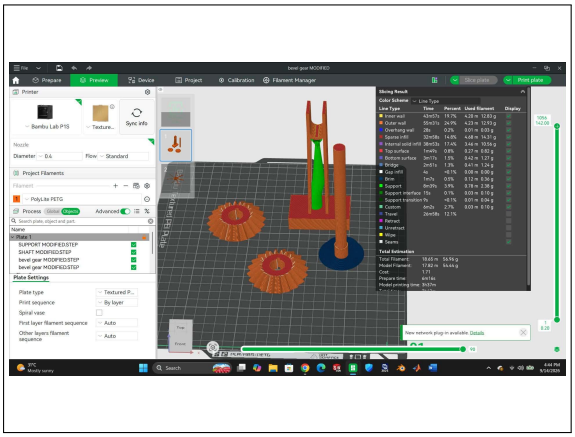
Prototype Assembly

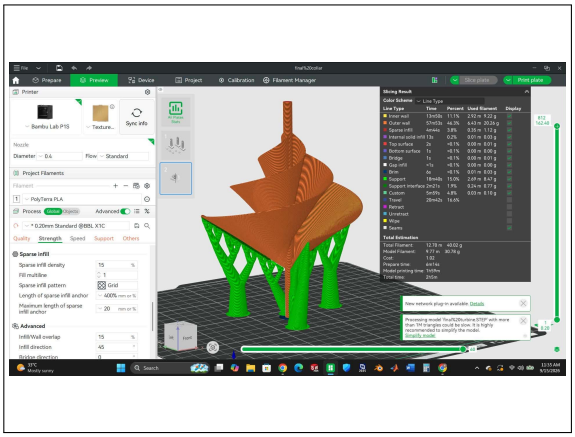


Technical Drawing

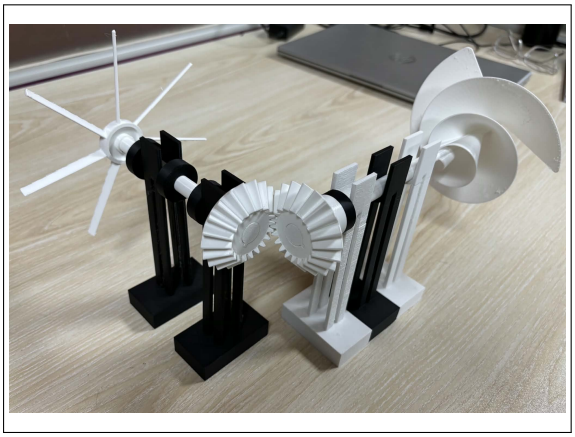
Project's 3D Printing Phase:

Screenshots of the project's slicing software





Images of the project 3D printer printing phases and the final product





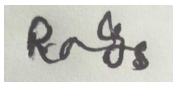




Additional Information:

Group Member 1

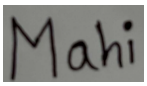
Name: Mohammad Rafiul Islam
Date: 15-09-2026



Signature: .....

Group Member 2

Name: Tasnia Basharat Mahi
Date: 15-09-2026



Signature: .....

2- Video Submission

Video Link

https://youtu.be/nL_0LNzpgFc?si=8fEcUzKmaCxL9pPB